Q1) Use mathematical induction to show that `MysteryFunction(n)` returns $-5n + \frac{(152)2^{2n+1}}{3} - \frac{31}{3}$, for n ≥ 1.

```
MysteryFunction (positive integer n)

    i = 1
    j = 20
    while i ≠ n + 2 do
        i = i + 1
        j = 4j + 15i - 19
    end while

return j
```

1. Base Case n = 1

For n = 1, while loop runs twice. In the first iteration, `i=1+1=2` and `j=4·20+15·2-19=91`. In the second iteration `i=2+1=3` and `j=4·(91)+15·3-19=390`. The function returns then j_1= `390`.

Formula evaluates $j_1 = -5\cdot(1) + \frac{(152)2^{2\cdot1+1}}{3} - \frac{31}{3} = 390$

Base case holds ✓

2. Use (given that) for n = k

Function returns $j_k = -5k + \frac{(152)2^{2k+1}}{3} - \frac{31}{3}$

3. Target for n = k +1

Function returns $j_{k+1} = -5(k+1) + \frac{(152)2^{2(k+1)+1}}{3} - \frac{31}{3} = -5k - 5 + \frac{(152)2^{2k+3}}{3} - \frac{31}{3} = -5k + \frac{(152)2^{2k+3}}{3} - \frac{46}{3}$

4. Proof

When n = k +1 is passed to the function, in the last iteration of the while loop j_{k+1} is calculated as:

$j_{k+1} = 4j_k + 15i - 19$. The condition `i ≠ n + 2` is `i ≠ k + 3`, meaning `i=k+2` and then incremented to `i= k + 3` in the next line. Therefore,

$j_{k+1} = 4j_k + 15(k+3) - 19$

$j_{k+1} = 4j_k + 15(k) + 45 - 19$

$j_{k+1} = 4j_k + 15(k) + 26$

From the Step 2 we have $j_k = -5k + \frac{(152)2^{2k+1}}{3} - \frac{31}{3}$ to substitute

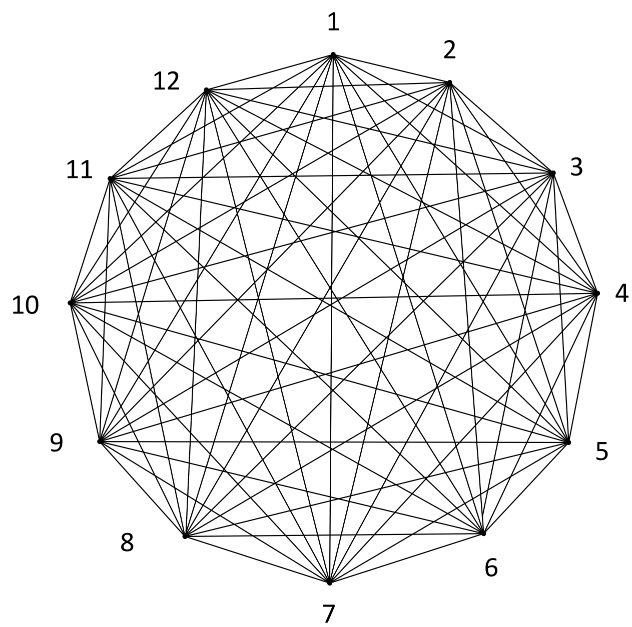
$j_{k+1} = 4\left(-5k + \frac{(152)2^{2k+1}}{3} - \frac{31}{3}\right) + 15(k) + 26 = -20k + 4 \cdot \frac{(152)2^{2k+1}}{3} - 4 \cdot \frac{31}{3} + 15k + 26$

$j_{k+1} = -20k + 2^2 \cdot \frac{(152)2^{2k+1}}{3} - \frac{124}{3} + 15k + 26 = -5k + \frac{(152)2^{2k+1+2}}{3} - \frac{46}{3}$

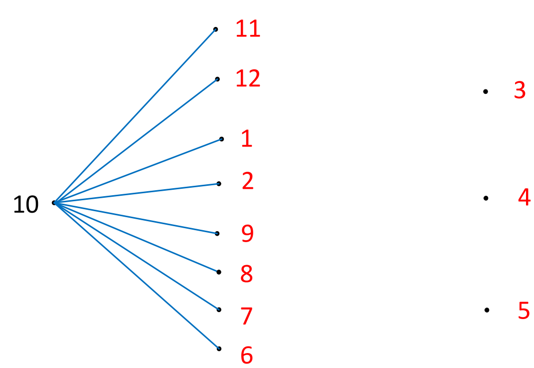
$j_{k+1} = -5k + \frac{(152)2^{2k+3}}{3} - \frac{46}{3}$ which matches the target.

∴ By the principle of mathematical induction, `MysteryFunction` returns $j_n = -5n + \frac{(152)2^{2n+1}}{3} - \frac{31}{3}$ for $n \geq 1$.

Q2) Consider the graph below, where every two vertices are connected. Find the number of spanning trees having the vertex 10 with a degree 8.



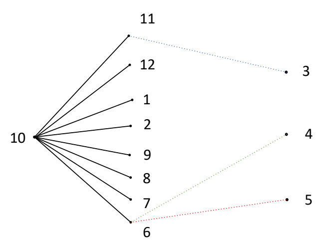
Let vertex 10 be the vertex connected to 8 other vertices. There are 11 remaining vertices, so this can be done in $\binom{11}{8}$ different ways. The following is one of those possibilities:



There is no edge connecting any pair of vertices 11, 12, 1, 2, 9, 8, 7, 6 as that would imply a cycle (tree has no cycle).

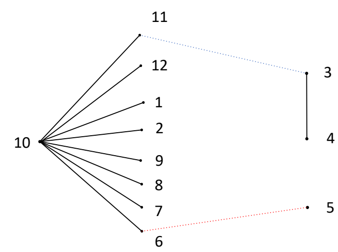
For the same reason, vertices 3, 4, 5 cannot be fully connected to each other (it makes a cycle).

To connect vertices 3, 4 and 5 to the existing tree, consider that vertices 3, 4 and 5 have a) no edge among them, or b) have one edge among them, or c) have two edges among them.



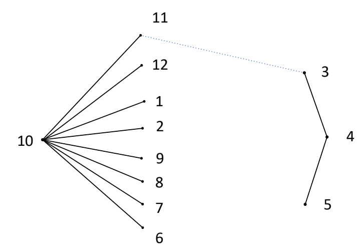
$8 \cdot 8 \cdot 8$ choices to connect vertices 3, 4, 5

OR



$\binom{3}{1} 8 \cdot 8 \cdot \mathbf{2}$ choices to connect vertices 3 and 5 **or** 4 and 5
$\binom{3}{1}$ is to select one solo vertex

OR



$\binom{3}{1} 8 \cdot \mathbf{3}$ choices to connect vertex 3 **or** 4 **or** 5
$\binom{3}{1}$ is to select one vertex with deg 2

Hence, the number of trees having vertex 10 with degree 8 is $\binom{11}{8}\left(8 \cdot 8 \cdot 8 + \binom{3}{1} 8 \cdot 8 \cdot \mathbf{2} + \binom{3}{1} 8 \cdot \mathbf{3}\right)$.

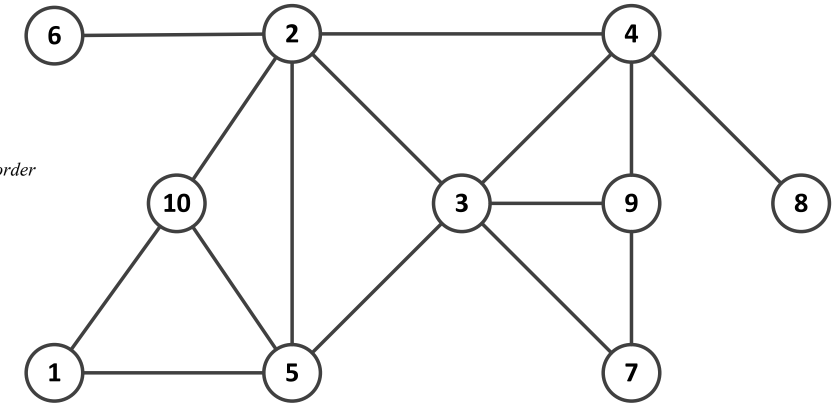
1) provide the final formula

$$\binom{11}{8}\left(8 \cdot 8 \cdot 8 + \binom{3}{1} 8 \cdot 8 \cdot 2 + \binom{3}{1} 8 \cdot 3\right)$$

2) evaluate the formula

There are $\frac{11 \cdot 10 \cdot 9}{3 \cdot 2 \cdot 1} \cdot (8 \cdot 8 \cdot 8 + 3 \cdot 8 \cdot 8 \cdot 2 + 3 \cdot 8 \cdot 3) = 165 \cdot (512 + 384 + 72) = 159{,}720$ spanning trees having the vertex 10 with a degree 8.

Q3) Run modified DFS algorithm starting at **vertex 5**.




```
dfs(vertex v)
  {
     for each neighbor w of v  //in the increasing order
        if w is unvisited
        {
        visit(w);
        print v
        dfs(w);
        print w
        }
  }
```

Show the following and delete empty cells at the end. Print the result in color, v – red, w - blue:

1) RETURN STACK

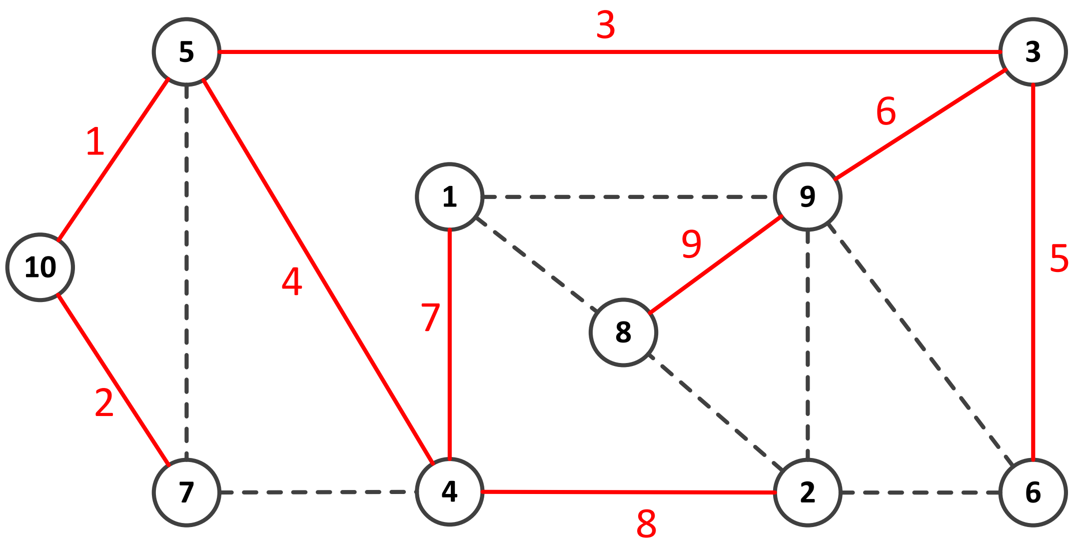
2) PRINTED ON THE SCREEN:

2
2
9
4
4
3
2
5
1
5

5	1	5	2	3	4	8	4	9	7	9	4	3	2	6	2	10	2	5	1

Q4) Run BFS algorithm starting at **vertex 10** and show its spanning tree. Give a label to each edge according to the order it is added to the spanning tree (i.e. edge that is added first label 1, edge that is added second label 2, etc.). Show queue Q and delete empty cells at the end.

Instructions: You must use algorithm presented in the class. Yes, you can label and highlight the edges by hand. Alternately, import the figure in Visio and label and highlight the edges – then paste it back into the document.



Queue Q:

10	5	7	3	4	6	9	1	2	8

Q5) We define "SomeOrder" tree Traversal Algorithm as follows:

SomeOrder Traversal Algorithm:

Step 1: (start)

Go to the root.

Step 2: (go right)

Go to the **right** subtree, if one exists, and do a SomeOrder traversal.

Step 3: (go right)

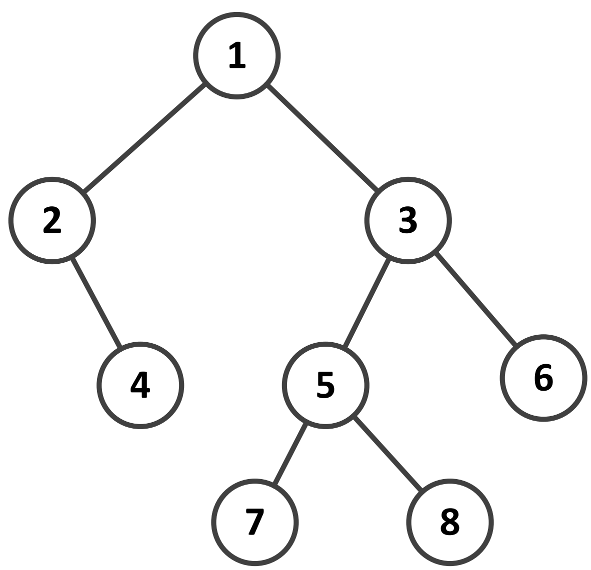
Go to the **right** subtree, if one exists, and do a SomeOrder traversal.

Step 4: (go left)

Go to the **left** subtree, if one exists, and do a SomeOrder traversal.

Step 5: (visit)

Visit the root.



For the tree shown, list the vertices according to the SomeOrder traversal.

6	6	8	8	7	5	3	6	6	8	8	7	5	3	4	4	2	1